



Amazon Listing Optimizer Agent

An AI agent that analyzes weak Amazon product listings and rewrites them to maximize clicks and conversions — using prompt chaining and structured LLM outputs.

Built this because: I was exploring how AI agents can automate real e-commerce pain points. Amazon sellers lose thousands of dollars to bad listings. This agent fixes that in seconds.

What it does

Raw weak listing → [Agent Step 1: Analyze] → [Agent Step 2: Optimize] → [Agent St

Step 1 — Analyze: Scores the existing listing (1–10), identifies 3 specific weaknesses, finds the top missing keyword

Step 2 — Optimize: Rewrites title, 5 bullet points, description, and generates 8 backend keywords

Step 3 — Score: Estimates CTR lift % and conversion lift % from the improvements

Demo output

```
=====
  AMAZON LISTING OPTIMIZER AGENT
=====
Product: Water Bottle Stainless Steel...

[Agent Step 1] Analyzing current listing...
  → Score: 4/10
  → Weaknesses found: 3

[Agent Step 2] Generating optimized listing...
  → New title length: 187 chars
  → Keywords generated: 8

[Agent Step 3] Calculating improvement metrics...
```

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RESULTS

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Score: 4/10 → 8/10



Estimated CTR lift: +12.8%



Estimated conversion lift: +7.2%



NEW TITLE:

Premium Stainless Steel Water Bottle 32oz | BPA-Free Insulated...



NEW BULLETS:

- STAYS COLD 24 HOURS: Double-wall vacuum insulation...
- NEVER LEAK AGAIN: Triple-seal lid technology...
- ...



Full output saved to output.json

Run it

```
# Install dependencies
pip install -r requirements.txt

# Set your Anthropic API key
export ANTHROPIC_API_KEY=your_key_here

# Run with demo listing
python amazon_listing_agent.py

# Run with your own listing
python amazon_listing_agent.py sample_listing.json
```

No API key? It runs in demo mode with realistic mock outputs — no setup needed.

Tech

- Python 3.9+
- Anthropic Claude API (claude-opus-4-5)
- Prompt chaining (multi-step agentic workflow)

- Structured JSON outputs
 - Zero external dependencies beyond `anthropic`
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Why I built this

I'm deeply interested in how AI agents can replace entire workflows — not just answer questions, but *do things*. Amazon listing optimization is a \$5,000/agency problem. This agent does it in under 30 seconds.

This is directly relevant to how I think about building at early-stage AI startups: find a real pain, build an agent, ship it.

Author

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